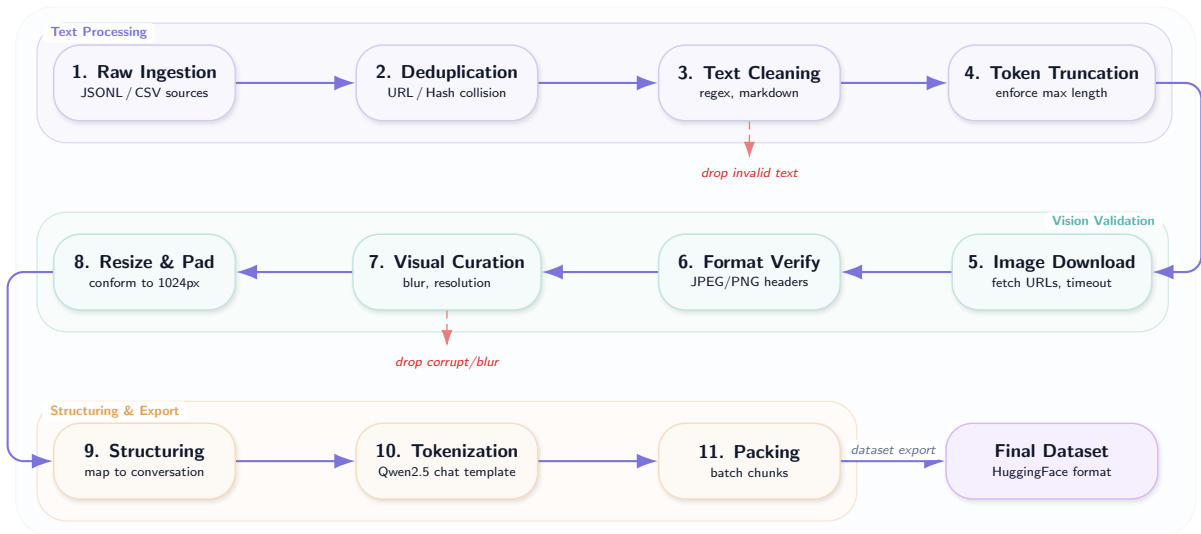


# VLM Data Pipeline Flow (11 Stages)

The serpentine sequence of ingestion, text cleaning, visual curation, and formatting into Qwen-style conversational arrays.



**Design rationale** — The pipeline is strictly sequential. Text cleaning occurs before image downloading to save bandwidth by immediately dropping malformed textual records. Visual curation guarantees that the model receives only crisp, adequately sized imagery, which is critical for the visual token representations in the Qwen architecture.